SAGoG: Similarity-Aware Graph of Graphs Neural Networks for Multivariate Time Series Classification

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***Abstract*—Multivariate Time Series Classification (MTSC) has important research significance and practical value. Deep learning models have achieved considerable success in addressing MTSC problems. However, a key challenge faced by existing classification models is how to effectively consider the correlations between time series instances and across channels simultaneously, as well as how to capture the dynamic of these inter-channel correlations over time. Current methods often fall short in these aspects: on one hand, they fail to fully account for the combined effects of inter-instance and inter-channel correlations; on the other hand, they largely overlook the dynamic nature of how inter-channel correlations change over time. To address these issues, we propose a novel graph neural network model, called Similarity-Aware Graph of Graphs neural networks (SAGoG), for multivariate time series classification. This model can comprehensively consider the dependencies between channel-level and instance-level time series, it dynamically learns dependency features through graph structure evolution and graph pooling layers. We conduct experiments on the UEA dataset to validate the SAGoG model, and the results demonstrate its outstanding performance in multivariate time series classification tasks.**

***Index Terms*—Multivariate time series, graph neural network, time series classification.**

I. Introduction

MULTIVARIATE Time Series (MTS) are widely available in many fields such as action recognition [1], healthcare [2], traffic management [3] and energy management [4], providing indispensable support for data analysis and decision-making in real-world scenarios. The continuous growth in data volume and the expansion of application scenarios underscore the increasing importance of research in time series data mining. Among these tasks, Multivariate Time Series Classification (MTSC) holds considerable research significance and practical value within the realm of time series data mining. MTSC methods aim to label multivariate time series with distinct categories, aiding in the comprehension of observed events and facilitating corresponding judgments. Given the increasing complexity of practical application scenarios, MTSC presents extensive potential for application across diverse domains.

MTSC has garnered widespread attention among researchers, primarily consisting of two categories of methods: methods based on distance measurement and methods based on deep learning. Historically, approaches based on distance measurement for computing similarity between sequences have dominated time series classification, centered on constructing a k-Nearest Neighbor (k-NN) classifier using similarity metrics [5]. Dynamic Time Warping (DTW) [6] stands out as one of the most popular similarity measures, employed to calculate the optimal alignment between time series. Existing studies suggest that combining DTW with classifiers leads to significant improvements in classification performance [7], [8], [9]. However, similarity measure is typically conducted in an unsupervised paradigm, often with a simplistic k-NN classifier as the backbone model, which may not ensure optimal classification results.

Deep learning methods model the feature representation of multivariate time series data through a unified process, eliminating the need for complex feature engineering. Fully Convolutional Networks (FCN) have distinct advantages over methods based on DTW in terms of time series classification results, demonstrating superior classification performance compared to traditional TSC algorithms [10], [11]. Furthermore, some methods are specifically designed to learn the latent features of MTS for the classification task. For instance, approaches like SIMTSC [12] and LB-SIMTSC [13] combine Graph Convolutional Networks (GCN) with DTW to accomplish classification tasks. These deep learning-based frameworks demonstrate promising results in MTSC tasks. However, the TSC models based on deep learning frequently overlook the implicit correlations between different samples and channels, which can limit their effectiveness.

Graph structures are highly suitable for modeling correlations. In the past few years, Graph Neural Networks (GNNs) have been widely applied with great success in the field of graph data processing due to their excellent ability to learn graph structures. By propagating various structural information within Graph Neural Networks, the representation of nodes is continually updated by aggregating information from neighboring nodes. As

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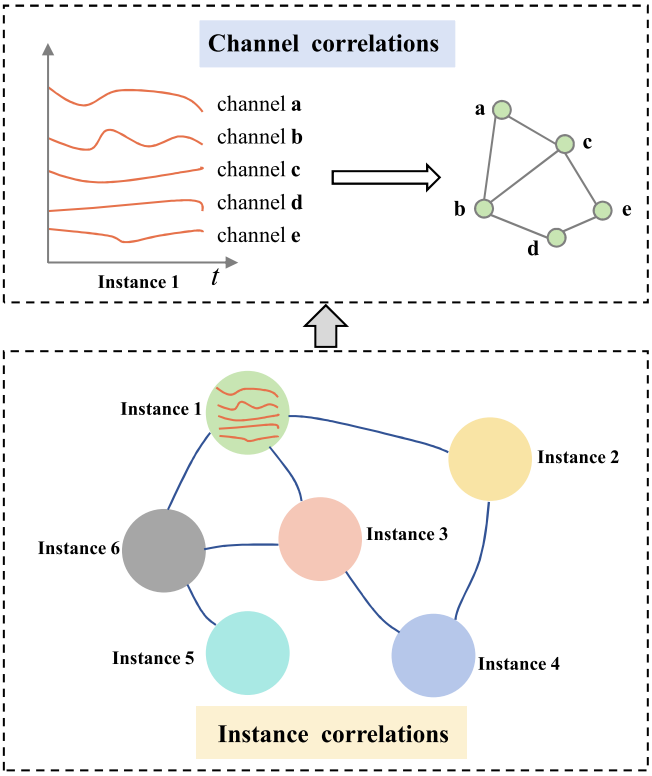

Fig. 1. Instance and Channel Correlation of Multivariate Time Series. Below the image, the inter-instance correlations are depicted via a graph structure. Each instance corresponds to a multivariate time series sample. Above the image, the inter-channel correlations are illustrated, with each channel representing a unit time series within an instance.

shown in Fig. 1, each sample or channel can be regarded as a node in the graph of MTS, interconnected through latent dependency relationships. Therefore, utilizing graphs to model MTS data offers significant potential. Existing studies [12], [13] have gradually endeavored to integrate Graph Neural Networks into MTSC. However, these methods usually adopt graph structures to model the correlations between different channels. Each MTS can be regarded as an instance, and the exploration of correlations between different instances is often overlooked. Different time series instances often exhibit shared patterns that cannot be observed from individual sequences alone. For example, similar pathological features may appear in ECG data from different patients, or common fault patterns may be present in data from various machines. Modeling such correlations can improve the generalization capability of the model and enhance classification performance. Additionally, for time series data, the dependency relationships between channels dynamically change. Existing methods do not adequately capture these evolving relationships.

To solve the aforementioned problems, we introduce an innovative Similarity-Aware Graph of Graphs neural networks (SAGoG) for MTSC. Unlike existing methods that focus solely on either channel-wise or instance-wise relationships, SAGoG integrates both perspectives within a unified framework. By dynamically evolving the channel graph and constructing an instance graph, our model captures both local feature interactions and global contextual correlations, leading to improved classification performance. To model channel-level and instance-level dependencies in time series, we employ three modules to learn the dependency features between channels and instances: Instance Graph Construction, Channel Graph Structure Evolving, and Channel Graph Pooling. The Instance Graph Construction module utilizes the DTW algorithm, while channel graph structures are learned from data embeddings. To learn the evolving dynamics in channel correlations, the Channel Graph Structure Evolving module combines recurrent neural networks with GNNs to obtain feature representations of time series data. In general, the main contributions of the SAGoG model are as follows:

1) We investigated the utilization of graph structures to model the dependencies between instances and channels in MTS. The introduction of graph structures enables us to comprehensively capture the complex correlations within time series data.
2) We introduced the Similarity-Aware Graph of Graphs neural networks. This model can capture the dynamic changes in channel-level correlations while simultaneously considering the similarity information between different instances, leading to a more accurate learning of feature representations for multivariate time series classification.
3) Our approach achieved outstanding classification performance on MTS datasets. This demonstrates that our proposed model exhibits excellent classification performance across various data scenarios.

The organization of this paper is as follows. Section I is the Introduction, which provides an overview of the main background and issues in MTSC, highlighting the key contributions of our model. Section II systematically reviews related work on MTSC and GNNs. Section III serves as preparatory work, introducing the main concepts and symbol definitions related to the research problem. Section IV details the specific aspects of the designed SAGoG model, explaining the composition and roles of different modules. Section V covers Experimental Setup and Results Analysis, presenting the experimental settings, baseline models, datasets, and analysis of experimental results. The final part is the Conclusion, summarizing the entire paper.

II. Related Works

In recent research, the MTSC problem has attracted significant attention, emerging as a critical topic in time series data mining. In this section, we summarize the latest advancements in MTSC and GNNs in modeling time series data.

A. Multivariate Time Series Classification

MTSC methods can be broadly classified into three main types: methods based on distance metrics, methods relying on sequence features, and methods leveraging deep learning techniques. Among these, distance-based approaches often combine the 1-nearest neighbor algorithm [14] with functions for computing distances between sequences to achieve classification. Notably, DTW stands out as a highly effective distance metric method, attracting significant attention from researchers in the

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field of MTSC [15], [16]. Methods based on sequence features achieve classification by exploring various characteristics inherent in the time series themselves. Shapelets, as a popular approach in this category [17], [18], can identify important subsequences relevant to class information. Zalewski et al. proposed a symbolic model for time series classification constructed using shapelet transformation [19]. WEASEL+MUSE [20] extracts discrete features from each window and dimension of multivariate time series, continuously encoding contextual information to extract and filter multivariate information. These two types of methods belong to traditional time series classification models. Although these methods have yielded effective results in MTS classification, they often rely on complex data processing and feature extraction tasks. Furthermore, these methods mainly extract shallow features of the time series, making it challenging to effectively capture deep-level sequence features and dependencies between multivariate variables.

Recently, deep learning methods exhibit excellent classification performance in addressing MTSC problems. MC-DCNN [21] proposed an effective Multi-Channels Deep Convolutional Neural Network architecture to learn features from multivariate time series data. Multivariate LSTM-FCNs [22] enhanced their fully convolutional blocks with squeeze-and-excitation blocks to further improve classification accuracy. TapNet [23] introduced a novel MTSC model with an attentional prototype network, aiming to integrate the strengths of traditional methods and deep learning-based approaches. MF-Net [24] proposed a multi-feature-based network to integrate local and global features of multivariate time series. These methods still primarily focus on leveraging deep learning techniques to represent sequential features of MTS data. However, for multivariate time series, spatial dependencies also play a crucial role in classification tasks. Graphs, as effective structures for representing non-euclidean relationships, are commonly used for exploring spatial correlations. GNNs have garnered attention in MTSC research. CauGNN [25] employed graph structures to represent multivariate time series. TodyNet [26] designed a novel temporal dynamic graph neural network to extract hidden spatiotemporal dynamic dependencies, enhancing classification accuracy. In these models, nodes in the graph represent variables, and edges represent correlations between variables. SIMTSC [12] and LB-SIMTSC [13] utilized graph neural networks to learn dependencies between different entities in MTS datasets, transforming the problem into a node classification task on the graph. MTSGNN [27] designed a novel graph neural network-based model for MTS classification that incorporates graph learning and interaction modules to capture dynamic relationships and improve classification performance.

As an excellent sequence modeling method, Transformer has shown outstanding performance in the field of time series analysis. Informer introduced the ProbSparse self-supervised mechanism to address long sequence prediction problems. Autoformer [28] combines time series decomposition strategies with the transformer framework, achieving good results in time series prediction. Informer [29] In recent years, large language models have introduced new solutions for time series prediction and classification problems [30]. Promptcast [31] and TIME-LLM [32] have designed time series prediction methods based on large language models, demonstrating excellent performance, particularly in zero-shot learning. Therefore, large language models also hold potential for time series classification tasks.

The above deep learning-based methods, particularly those utilizing GNNs, have shown significant advancements in learning features of MTS data. However, these approaches typically focus on exploring dependencies at either the channel or instance level independently, without simultaneously leveraging both sets of information.

B. Graph Neural Networks for Time Series

Recently, there has been a rising interest in using GNNs for time series analysis, attributed to their capability to effectively model the complex relationships and dependencies in time-varying data. GNNs have shown strong performance across various time series analysis tasks, including but not limited to classification [13], [26], prediction [33], [34], and anomaly detection [35], [36], [37]. Existing studies have explored various architectures and methods to leverage graph networks for encoding spatial information of time series data. For instance, some studies [37], [38] have proposed spatiotemporal graph structures to learn spatial correlations among different time points and the temporal dependencies within each time point. MT-GNN [39] employed graph learning modules to automatically extract directional relationships among variables for time series prediction. MAGNN [40] designed a multimodal GNN layer to learn financial time series prediction from these multimodal inputs. METRO [33] designed a multi-scale spatiotemporal graph neural network framework, which can simultaneously model dynamic and cross-scale variable correlations. MST-GAT [41] combined multimodal graph attention networks and temporal convolutional networks for anomaly detection. ForecastGrapher [42] reframes multivariate time series forecasting as a node regression task, using a Group Feature Convolution GNN to capture inter-series correlations and temporal dynamics effectively. TISER-GCN [43] is a new graph neural network designed for long multivariate time series regression, improving predictions using spatial and sequential data.

Graph neural network-based models have garnered significant attention in time series analysis. In our model, we use graph structures to model the correlations between multiple time series channels and instances, aiming to achieve more precise classification results.

III. Preliminaries

In this section, we begin by outlining the primary definition of the MTSC problem, followed by an in-depth exploration of the key definitions and symbols employed throughout the paper.

A. Problem Formulation

The main goal of this paper is to design a classifier that efficiently handles multivariate time series data, denoted as

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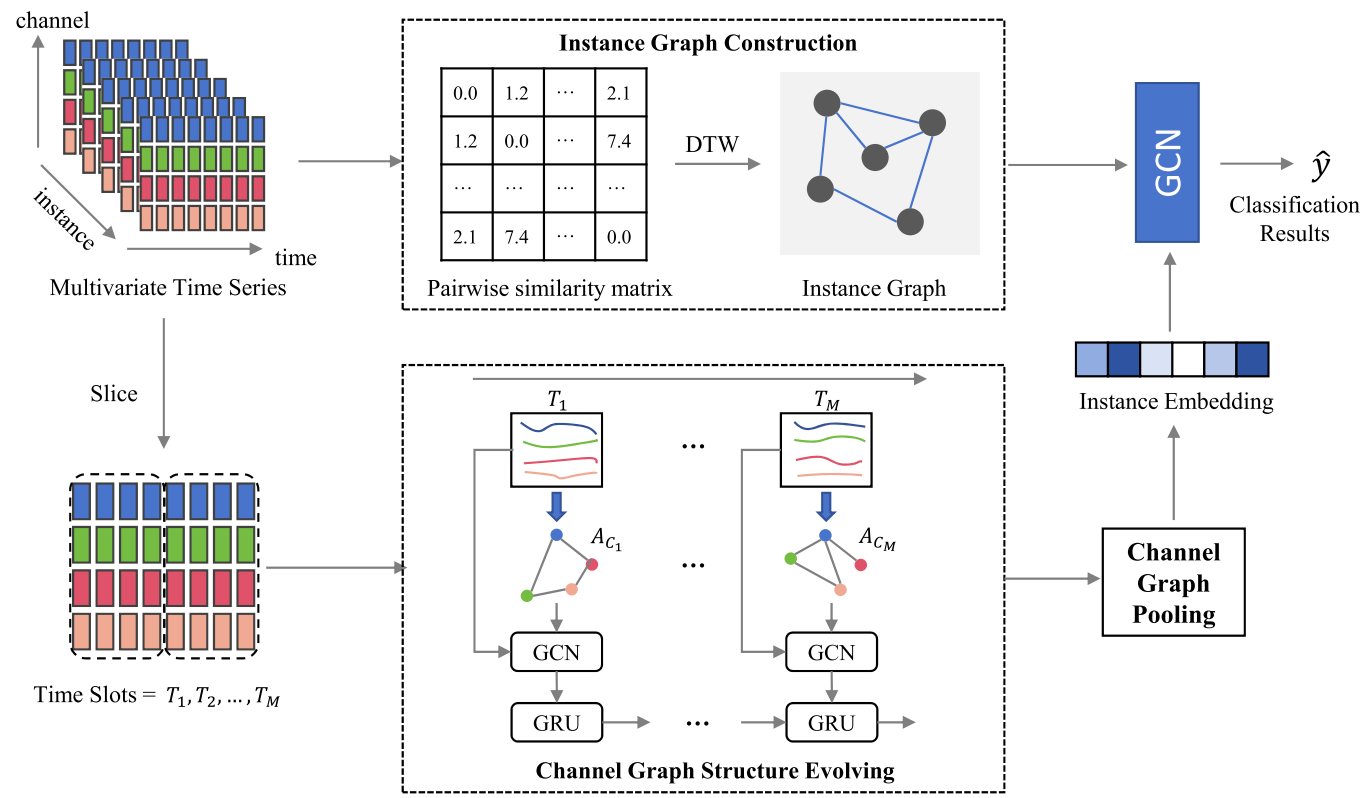

Fig. 2. The overall framework of SAGoG model. Our model consists of four main components: Instance Graph Construction, Channel Graph Structure Evolving, and Channel Graph Pooling.

$\mathbf{Z} = \{\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_n\} \in \mathbb{R}^{n \times t}$. Here, each $\mathbf{z}_j$ represents a univariate time series, reflecting the variations of the jth variable over time and is considered a channel in the model. The MTS dataset can be presented as $\mathbf{\Theta} = \{\mathbf{Z}_1, \mathbf{Z}_2, \ldots, \mathbf{Z}_k\}$, where it encompasses a collection of k distinct MTS. We refer to each MTS $\mathbf{Z}$ as an instance. Each instance belongs to a specific category, matching a label l from the set $\Lambda = \{l_1, l_2, \ldots, l_k\}$. The symbol k represents the number of time series categories contained in the MTS dataset. The objective of our study is to develop a mapping function $g(\cdot)$ that accurately associates each time series from the MTS dataset $\mathbf{\Theta}$ with the corresponding label from the set Λ.

B. Primary Symbols and Definitions

In the domain of multivariate time series classification, the utilization of graph structures provides a profound method for elucidating the intricate relationships among diverse entities. Within this context, we introduce concepts of the Instance Graph and Channel Graph, extending to the dynamic aspect with Evolving Channel Graph, to capture the essence of temporal and spatial relationships inherent in multivariate data.

Instance Graph: An Instance Graph is used to represent relationships between multiple MTS instances. The formal definition of an instance graph is given by $\mathbf{G}_I = (\mathbf{V}_I, \mathbf{E}_I)$, where $\mathbf{V}_I = \{\mathbf{v}_1, \ldots, \mathbf{v}_m\}$ consists of vertices, with each vertex representing a distinct MTS. The edge set $\mathbf{E}_I = \{(\mathbf{Z}_i, \mathbf{Z}_j)\}$ describes the relationships between these MTS (e.g., similarity or structural connections). Here, m represents the number of instances in the dataset, i.e., the count of MTS.

Channel Graph: The Channel Graph highlights interactions between its constituent univariate time series, revealing the internal relationships within each multivariate time series. Each MTS $\mathbf{Z}$ can be decomposed into a collection of multiple univariate time series$\{\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_n\}$. For a given multivariate time series Z, the channel graph is denoted as $\mathbf{G}_C = (\mathbf{V}_C, \mathbf{E}_C)$. Here, $\mathbf{V}_C = \{\mathbf{v}_{C1}, \ldots, \mathbf{v}_{Cn}\}$ represents nodes, with each node corresponding to a univariate component $\mathbf{z}$. The edge set $\mathbf{E}_C = \{(\mathbf{v}_{Ci}, \mathbf{v}_{Cj})\}$ captures intricate relationships between these univariate time series.

Evolving Channel Graph: The evolving channel graph emphasizes the dynamic nature of relationships within multivariate time series by partitioning them into discrete time segments (denoted as $T_1, T_2, \ldots, T_M$). This approach illustrates the temporal evolution of these interactions. Based on the data segmented into time periods $m = \{1, \ldots, M\}$, we construct a series of channel graphs $\mathbf{G}_C^m = (\mathbf{V}_C, \mathbf{E}_C^m)$. Each $\mathbf{E}_C^m$ is generated according to the data of its respective time segment, reflecting the channel correlations within the specified time period T_m.

IV. Methodology

As shown in Fig. 2, our model is composed of four components: Instance Graph Construction, Channel Graph Structure Evolving, Channel Graph Pooling, and Graph-based Time Series

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Classification. We will provide a detailed explanation of the specific composition and functions of these four parts.

The purpose of constructing instance graphs is to create a graph structure that represents the relationships among all time series instances in the dataset. The channel graph evolving module learns the channel graph structures over different time periods of the time series. As time progresses, the channel graph structures continually change, and this module is capable of capturing the evolution of these structures. The channel graph pooling module aggregates the overall feature representation of each time series instance. The graph-based time series classification module learns the representations of time series instances based on the instance graph structure, ultimately producing the results of time series classification.

A. Instance Graph Construction

Due to the absence of explicit relational graphs in time series datasets, we adopt a similarity-based approach to construct instance graph structures for multivariate time series. Similar to SIMTSC [12] and LB-SIMTSC [13], we employ the DTW algorithm, which serves as an elastic distance measure for quantifying similarity between different time series.

Each multivariate time series instance in this study is represented as $\mathbf{Z} \in \mathbb{R}^{n \times l}$, composed of univariate time series from different channels $\{\mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3, \ldots, \mathbf{z}_n\}$. To calculate the similarity between multivariate time series instances, we preprocess them into one-dimensional time series $\hat{\mathbf{Z}}$ to facilitate the application of the Dynamic Time Warping (DTW) algorithm. The purpose of this flattening is to simplify instance similarity calculations, leveraging DTW's capability to find optimal matching paths between sequences. By constructing a dynamic programming matrix between the two sequences, the optimal matching path can be computed, considering the relative relationships between points in the sequences and allowing for non-linear alignment of the time axes. This process of constructing the optimal matching path is called time warping, and the resulting warped distance is the DTW distance [6].

Based on the obtained time alignment path, we can construct the instance graph structure for MTS. Suppose the MTS of the input is represented as $\mathbf{\Theta} = \{\mathbf{Z}_1, \mathbf{Z}_2, \ldots, \mathbf{Z}_k\}$. By applying the aforementioned algorithm to the multivariate time series data $\mathbf{\Theta}$, we can obtain instance correlations between different time series, forming a instance graph $\mathbf{G}_I = \{\mathbf{A}_I, \mathbf{E}_I\}$ that reflects the relationships between the sequence instances. Let $D(i, j)$ denote the DTW distance between two MTS instances $\hat{\mathbf{Z}}_i$ and $\hat{\mathbf{Z}}_j$. We construct the adjacency matrix $\mathbf{A}_I$ of the instance graph using the DTW distance, defined as follows:

$$\mathbf{A}_I^{ij} = \frac{1}{\exp(\alpha D(i, j))}, \tag{1}$$

where $\mathbf{A}_I^{ij}$ represents the correlation between instances i and j, and α is a scaling parameter. In this graph, nodes denote different instances of MTS data, and the edge weights represent the similarity measure, specifically the DTW distance.

B. Channel Graph Structure Evolving

Due to the dynamic nature of correlations between channels in multivariate time series, leveraging this correlation information in a time series classification model presents a challenge. To address this issue, we introduce a channel graph structure evolving module designed to learn the evolving channel correlations over time. This module consists of channel graph generation, graph convolutional layers, and GRU layers.

Initially, we generate the initial adjacency matrices $\mathbf{A}_C$ for different time segments based on the temporal data. A MTS instance $\mathbf{Z}$ is composed of multiple univariate time series $\{\mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3, \ldots, \mathbf{z}_n\}$. In this context, we employ a data-driven approach to generate the channel graph structure $\mathbf{G}_C$ for the multivariate time series. This method differs from similarity-based approaches as it relies on the embedded representations of the time series to learn the channel adjacency matrix. Moreover, the relationships between the individual time series are not constant over time [26]. Therefore, we uniformly divide the input time series into M segments based on a specified time length. Assuming the total time length is T, the segmented time periods are denoted as $\mathbf{T} = \{T_1, T_2, \ldots, T_M\}$. Based on these segmented data, we generate the channel graph structures $\mathbf{G}_C = \{\mathbf{G}_C^1, \mathbf{G}_C^2, \ldots, \mathbf{G}_C^M\}$ for different time periods.

During the construction of the channel graph structure, we initially create two embeddings $\mathbf{E}_1$ and $\mathbf{E}_2$ for each time period. These embeddings are parameters learned during the model training process and are initialized randomly. Subsequently, we use these embeddings to generate the initial adjacency matrix for the respective time period using the following formula:

$$\mathbf{A}_C = softmax(relu(\mathbf{E}_1^T \mathbf{E}_2)) \tag{2}$$

In our time series classification model, the adjacency matrix $\mathbf{A}_C$ undergoes iterative training to converge to a channel graph structure that better fits the actual data. The learned adjacency matrix is then sparsified to reduce computational complexity in subsequent message-passing steps. This involves selecting the top k values for each row or column of the matrix, setting these positions to 1, and the rest to 0. Finally, we obtain the channel graph structure matrices for the multivariate time series as $\mathbf{A}_C = \{\mathbf{A}_{C_1}, \mathbf{A}_{C_2}, \ldots, \mathbf{A}_{C_M}\}$.

Following this, GCNs are employed to learn the complex relationships between nodes in the channel graph structure. The GCNs extract high-level features of the channel graph through multiple convolution operations, thereby enhancing the understanding of dynamic relationships between channels. As the core component of this module, the graph convolutional layers perform convolution operations on the channel graph, integrating information from the adjacency matrices to capture abstract relationships between channels. Utilizing multiple layers of graph convolutions facilitates the extraction of deeper feature representations, enabling the model to comprehensively represent the channel graph structure of multivariate time series. Specifically, the adjacency matrix of the channel graph for time segment T_1 is $\mathbf{A}_{C_1}$, and the input trajectory representation is $\mathbf{X}_{T_1}$. The computation formula for the graph convolutional layer

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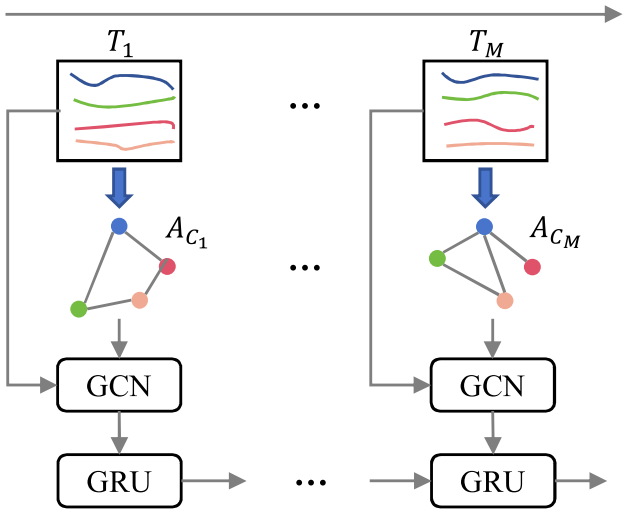

Fig. 3. This figure visually represents the sequential steps involved in the proposed methodology, emphasizing the adaptability of the model to evolving relationships within multivariate time series data.

is as follows:

$$f_c(\mathbf{X}_1) = \mathbf{A}_{C_1}\mathbf{X}_{T_1}\mathbf{W}_1 \tag{3}$$

The Channel Graph Structure Evolving module incorporates GRU layers, which are responsible for learning the dynamic features of the evolving channel graph over time. By learning the evolution patterns of the channel graph structure, the GRU layers can update the channel correlation weights across different time segments, enabling the model to flexibly adapt to the changing correlations between channels. The computation process of the GRU layers is as follows:

$$\mathbf{r}_c^t = \sigma(f_c([\mathbf{X}^t, \mathbf{H}^{t-1}], \mathbf{A}_c) + \mathbf{b}_r) \tag{4}$$

$$\mathbf{u}_c^t = \sigma(f_c([\mathbf{X}^t, \mathbf{H}^{t-1}], \mathbf{A}_c) + \mathbf{b}_u) \tag{5}$$

$$\mathbf{c}_c^t = tanh(f_c([\mathbf{X}^t, \mathbf{r}^t \odot \mathbf{H}^{t-1}], \mathbf{A}_c) + \mathbf{b}_c) \tag{6}$$

$$\mathbf{H}_c^t = \mathbf{u}^t \odot \mathbf{H}^{t-1} + (1 - \mathbf{u}^t) \odot \mathbf{c}^t \tag{7}$$

where $\mathbf{X}^t$ and $\mathbf{H}_c^t$ are the inputs and outputs of the GRU unit at moment t, and $\mathbf{r}_c^t$ and $\mathbf{u}_c^t$ denote the reset and update gates, respectively. $\mathbf{b}_r$, $\mathbf{b}_u$ and $\mathbf{b}_c$ are the deviation terms.

C. Channel Graph Pooling

In the SAGoG model, the MTSC problem is treated as a node classification task where nodes represent MTS instances, and the graph's adjacency matrix corresponds to the instance graph $\mathbf{A}_I$. For node classification, the first step involves obtaining embedding representations of nodes (MTS instances). Each MTS instance contains multiple channels representing univariate time series., which are represented using a channel graph structure. Thus, it is essential to derive the channel graph representation for each MTS instance.

Pooling is a critical step in obtaining the comprehensive channel graph representation. Throughout the evolution of the channel graph structure, dynamic dependencies among channels are learned. At this stage, the final output includes both time series features $\mathbf{H}_C^M$ and the adjacency matrix $\mathbf{A}_{C_M}$. In the SAGoG model, each instance $\mathbf{X}$ consists of various channels of single-variable time series $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \ldots, \mathbf{x}_n\}$. The relationships between instances are also depicted using the graph structure G_I. To effectively learn dependencies between instances using graph neural networks, the acquired time series features $\mathbf{H}_C^M$ initially undergo graph pooling to derive an overall representation of the instances. To preserve the information of the channel graph structure, hierarchical pooling methods are employed to obtain comprehensive representations of each instance.

Hierarchical pooling is performed on the channel graph structure by hierarchically aggregating nodes representing the time series of different channels. The channel graph is denoted as $G_C = (\mathbf{V}_C, \mathbf{E}_C)$, comprising n nodes. A key operation in hierarchical aggregation is learning the assignment matrices $\mathbf{S}$ at different levels. In each level, all nodes are mapped to several clusters, which then serve as input nodes for the next level. This hierarchical pooling process reduces the number of nodes as the hierarchy deepens, allowing for more abstract and high-level representations. At level l, the time series nodes are represented as $\mathbf{H}_C^l \in \mathbb{R}^{n_l \times d}$, where n_l is the number of nodes at this level and d is the feature dimension of each node. The assignment matrix $\mathbf{S}^{pool}$ at this level is computed as follows:

$$\mathbf{S}^{pool} = \text{softmax}(\sigma(\mathbf{A}_C^l\, \mathbf{H}_C^l\, \mathbf{W}_C^l)) \tag{8}$$

where $\mathbf{S}^{pool} \in \mathbb{R}^{n_l \times n_{l+1}}$ and $n_l > n_{l+1}$. Here, $\mathbf{A}_C^l \in \mathbb{R}^{n_l \times n_l}$ represents the node correlation matrix at level l, encoding the pairwise relationships between nodes. $\mathbf{W}_C^l \in \mathbb{R}^{d \times n_{l+1}}$ is a learnable parameter matrix that transforms node features to determine their cluster assignments. $\sigma(\cdot)$ is a non-linear activation function.

Through this operation, each node at level l is assigned to one of the n_{l+1} clusters at level $l+1$, ensuring structured information aggregation. Different from DiffPool [44], we employ a hard assignment strategy, meaning that each node belongs to exactly one cluster. The calculation of $\mathbf{S}^l$ is as follows:

$$\mathbf{S}_{ij}^{(l)} = \begin{cases} 1 & \text{if } \mathbf{S}_{ij}^{pool} = \max(\mathbf{S}_{i,:}^{pool}) \\ 0 & \text{otherwise} \end{cases} \tag{9}$$

where $\mathbf{S}_{i,:}^{pool}$ denotes the ith row of the matrix $\mathbf{S}^{pool}$, and $max(\mathbf{S}_{i,:}^{pool})$ is the maximum value in the ith row of $\mathbf{S}^{pool}$. After obtaining the assignment matrix, we can learn to obtain the node feature representation $\mathbf{X}^{l+1} \in \mathbb{R}^{n^{l+1} \times d^{l+1}}$ for the $l+1$ th layer and the adjacency matrix $\mathbf{A}_C^{l+1} \in \mathbb{R}^{n^{l+1} \times n^{l+1}}$. The formula is shown below:

$$\mathbf{X}^{l+1} = \mathbf{S}^{(l)^T}\mathbf{H}_C^l \tag{10}$$

$$\mathbf{A}_C^{l+1} = \mathbf{S}^{(l)^T}\mathbf{A}_C^l\mathbf{S}^{(l)} \tag{11}$$

After multiple layers of hierarchical pooling layers, the channel graph representation $\mathbf{C}_t$ is obtained for each time period. Next, we aggregate the learned hierarchical representations of the different evolved channel graphs to obtain the final representation of the entire multivariate time series: $\mathbf{H}'_C = \sigma(\sum_1^M \mathbf{C}_t\mathbf{W})$.

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D. Graph Based Time Series Classification

In the SAGoG model, $\mathbf{H}'_C$ represents the feature representation of each multivariate time series instance, while $\mathbf{A}_I$ denotes the adjacency matrix of the instance graph structure. Specifically, $\mathbf{H}'_C$ is the feature representation of multivariate time series instances obtained through channel graph pooling and hierarchical pooling. Meanwhile, $\mathbf{A}_I$ is the adjacency matrix of the instance graph structure, calculated using the DTW algorithm. In the instance graph node classification model, $\mathbf{H}'_C$ is used as the initial feature representation for each multivariate time series instance. The model construct a graph neural network model using the edge weight information from $\mathbf{A}_I$ and the node feature information from $\mathbf{H}'_C$. Then, multivariate time series classification is formulated as a node classification problem on an instance graph, constructed using DTW over the entire dataset. During training, the number of graph nodes equals the number of training samples, while in testing, it corresponds to the test samples. A Boolean mask ensures that the model learns only from training nodes and evaluates solely on test nodes.

This approach allows leveraging the information of nodes and edges in the instance graph structure to learn richer feature representations for each node. Subsequently, we perform graph convolution operations on $\mathbf{A}_I$ to facilitate information propagation between nodes and learn feature representations of multivariate time series instance nodes. Finally, by connecting a classifier layer composed of fully connected layers, we predict the classification labels for each node, thereby obtaining the classification results for multivariate time series. The calculation formula for the classification loss is as follows:

$$\mathcal{L} = -\sum_{i=1}^{N}\sum_{j=1}^{C} \left(y_{ij} log(\hat{y}_{ij}) - (1 - y_{ij}) log(1 - \hat{y}_{ij})\right), \quad (12)$$

where $\hat{y}_{ij}$ represents the probability value of prediction type. We adopt the multi-label classification focal loss as the model loss.

V. Experimental Results and Analysis

To verify the classification performance of SAGoG model, we conducted time series classification experiments on the UEA dataset. Moreover, a series of variants were designed in ablation experiments to evaluate the classification ability of the key components of the SAGoG model. All experiments were conducted on a server with a GPU 3090 (24 GB memory).

A. Datasets and Setup

The UEA dataset[1] is a popular MTS datasets used in current research. This dataset contains real-world MTS data obtained from various sources, including human activity recognition, audio sequence classification, and ECG/EEG signal classification. We selected 22 datasets from this collection. The length of these time series ranges from 8 to 17,984, and the dataset sizes vary from 27 to 10,992. Detailed information about the selected datasets can be found in Table I. In addition to the basic information of the 22 datasets, we also calculated the Gini coefficient [45] to measure the balance of category distribution within the datasets. Based on the Gini coefficient results, most datasets exhibit a balanced data distribution.

We evaluated the classification performance of the SAGoG model on 22 UEA datasets. The configuration of the train and test sets for different datasets can be found in Table I. The SAGoG model includes 4 dynamic channel graphs, with each time series divided into 4 time segments. The channel graph evolution module has 1 GCN layer, and the channel graph pooling module is configured with 2 pooling hierarchies. During the experiments, the batchsize was 16, and the learning rate and training epochs were 0.0001 and 2000. All experiments were conducted in the Pytorch 1.9 environment.

Evaluation Metric: To measure the performance of classification models, metrics such as accuracy, total best accuracy, average accuracy, and win/draw/loss counts in one-on-one comparisons are computed. Additionally, the critical difference diagram based on Wilcoxon signed-rank tests [24], [26] is constructed to assess the classification performance of SAGoG relative to other baseline models.

B. Baseline Models

We compare the SAGoG model with 9 baselines:

1) *ED-1NN [46]:* ED-1NN is a classical time series classification model.
2) *DTW-1NN-I [46]:* A commonly used baseline model for time series classification is the dimension-independent DTW method.
3) *DTW-1NN-D [46]:* The dimension-dependent DTW model, which is capable of processing all time dimensions simultaneously, offers a comprehensive approach to time series analysis.
4) *MLSTM-FCN [22]:* This model combines a squeeze-and-excitation block-enhanced FCN with a LSTM, effectively handling classification tasks.
5) *ShapeNet [47]:* The ShapeNet model obtains the embedding of shapelets and achieves MTSC by selecting key shapes.
6) *WEASEL+MUSE [20]:* A precise classifier is utilized to handle MTS data, involving feature extraction and selection across multiple dimensions, followed by decision-making using machine learning classifiers.
7) *TapNet [23]:* TapNet combines the strengths of traditional and deep learning methods, proposing a random group permutation approach to extract features from MTS data.
8) *OS-CNN and MOS-CNN [46]:* These models introduce improvements to 1D-CNNs used in time series classification tasks by automatically selecting kernel sizes that effectively capture optimal receptive field sizes.

[1] http://www.timeseriesclassification.com/

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TABLE I
DETAILS OF THE 22 DATASETS FOR MTSC

Datasets	Type	Series Number	Series Length	Classes Number	Train Size	Test Size	Gini Coefficient
ArticularyWordRecognition (AWR)	MOTION	9	144	25	275	300	0.00
AtrialFibrillation (AF)	ECG	2	640	3	15	15	0.00
DuckDuckGeese (DDG)	AUDIO	1345	270	5	50	50	0.00
EigenWorms (EW)	MOTION	6	17984	5	128	131	0.28
Epilepsy (EP)	HAR	3	206	4	137	138	0.04
FaceDetection (FD)	EEG	144	62	2	5890	3524	0.00
FingerMovements (FM)	EEG	28	50	2	316	100	0.00
HandMovementDirection (HMD)	EEG	10	400	4	160	74	0.05
Handwriting (HW)	HAR	3	152	26	150	850	0.08
Heartbeat (HB)	AUDIO	61	405	2	204	205	0.22
Libras (LI)	HAR	2	45	15	180	180	0.00
LSST (LS)	OTHER	6	36	14	2459	2466	0.54
MotorImagery (MI)	EEG	64	3000	2	278	100	0.00
NATOPS (NA)	HAR	24	51	6	180	180	0.00
PenDigits (PD)	MOTION	2	8	10	7494	3498	0.02
PEMS-SF (PE)	MISC	963	144	7	267	173	0.02
PhonemeSpectra (PS)	SOUND	11	217	39	3315	3353	0.00
RacketSports (RS)	HAR	6	30	4	151	152	0.05
SelfRegulationSCP1 (SCP1)	EEG	6	896	2	268	293	0.00
SelfRegulationSCP2 (SCP2)	EEG	7	1152	2	200	180	0.00
StandWalkJump (SWJ)	ECG	4	2500	3	12	15	0.00
UWaveGestureLibrary (UGL)	HAR	3	315	8	120	320	0.00

9) *Autoformer [28]:* Autoformer is a time series forecasting model based on a decomposition strategy, effectively capturing periodicity and trend features in long sequences.
10) *Informer [29]:* Informer efficiently handles long-sequence forecasting tasks by introducing the ProbSparse self-attention mechanism and a single-layer encoder-decoder structure.
11) *Promptcast [31]:* PromptCast is a novel time series forecasting paradigm based on large language models, transforming numerical inputs and outputs into prompts and reframing the forecasting task as a sentence-to-sentence generation problem.
12) *TISER-GCN [43]:* TISER-GCN is a graph neural network architecture designed for long time series, transforming time series analysis into a node regression task by constructing graph structures.
13) *ForecastGrapher [42]:* ForecastGrapher reconstructs multivariate time series forecasting as a node regression task using a graph neural network.
14) *TodyNet [26]:* TodyNet is an optimal MTSC model that utilizes GNNs to learn the dynamic spatiotemporal features of MTS through a dynamic graph structure.

C. Classification Results and Analysis

Table II presents a comparison of classification accuracy across 22 UEA MTS datasets for 9 baseline models and SAGoG model. Each dataset lists the classification accuracy for different algorithms, with key metrics including the highest accuracy (bolded), the second-best accuracy (underlined), and the average accuracy across all datasets for each algorithm. The baseline model results in the table are referenced from the respective papers or Omni-Scale CNNs [46]. These results provide a comprehensive evaluation of each algorithm's performance across diverse datasets.

In this experiment, our proposed SAGoG model demonstrated significant superiority compared to multiple baseline models on 22 UEA benchmark datasets. The SAGoG model achieved an average classification accuracy of 0.740, surpassing all other baseline models, including ShapeNet (0.698) and TodyNet (0.709). On specific datasets, the SAGoG model showed notable improvements in classification performance. For example, the SAGoG model achieved the highest classification accuracy on the DuckDuckGeese, FaceDetection, MotorImagery, and PhonemeSpectra datasets, highlighting its strong capability in handling diverse time series datasets. In terms of 1-to-1 comparisons, the SAGoG model also performed exceptionally well. The SAGoG model achieved at least 14 wins (1-to-1-Wins) across 22 datasets, significantly outperforming other well-performing models such as TapNet (16 wins), OS-CNN (14 wins), MOS-CNN (14 wins), and TodyNet (15 wins). Additionally, the SAGoG model had at most 8 losses/draws (1-to-1-Losses/Draws), further demonstrating its significant overall performance advantage.

From the analysis of the results in Table II, it is evident that the SAGoG model achieves higher classification accuracy than current state-of-the-art methods on multiple datasets. For instance, SAGoG improves accuracy by 4.1% on the PE dataset, by 3.6% on the FM dataset, and by 8.9% on the FD dataset. These experimental results demonstrate that the SAGoG model performs exceptionally well in multivariate time series classification. Across 22 UEA datasets, we achieved the best classification results on 12 datasets, including DDG, FD, FM, HB, LS, MI, NA, PE, PS, SCP1, SCP2, and SWJ. Among these, the DDG, FD, and PE datasets have the largest number of series, with 1345, 144, and 963 respectively, making them the three datasets with the most channels. This indicates that our model effectively learns the channel correlations in MTS data, thereby improving classification accuracy. Furthermore, the FD, FM, MI, SCP1, SCP2, and SWJ datasets are all EEG or ECG types. The experimental results further validate that the SAGoG model is capable of learning the inter-channel and inter-instance dependencies of these classic multivariate time series signals.

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TABLE II
PERFORMANCE COMPARISON ACROSS 22 UEA MTS DATASETS

Datasets	ED-1NN	DTW-1NN-I	DTW-1NN-D	MLSTM-FCN	ShapeNet	WEASEL+MUSE	TapNet	OS-CNN	MOS-CNN	TodyNet	SAGoG
AWR	0.970	0.980	0.987	0.973	0.987	0.990	0.987	0.988	**0.991**	0.987	0.946
AF	0.267	0.267	0.200	0.267	0.400	0.333	0.333	0.233	0.183	**0.467**	0.430
DDG	0.275	0.550	0.600	0.675	0.725	0.575	0.575	0.540	0.615	0.580	**0.790**
EW	0.550	0.603	0.618	0.504	0.878	**0.890**	0.489	0.414	0.508	0.840	0.815
EP	0.667	0.978	0.964	0.761	0.987	**1.000**	0.971	0.980	0.996	0.971	0.928
FD	0.519	0.513	0.529	0.545	0.602	0.545	0.556	0.575	0.597	0.627	**0.683**
FM	0.550	0.520	0.530	0.580	0.589	0.490	0.530	0.568	0.568	0.570	**0.610**
HMD	0.279	0.306	0.231	0.365	0.338	0.365	0.378	0.443	0.361	**0.649**	0.541
HW	0.371	0.509	0.607	0.286	0.451	0.605	0.357	0.668	**0.677**	0.436	0.643
HB	0.620	0.659	0.717	0.663	0.756	0.727	0.751	0.489	0.604	0.756	**0.761**
LI	0.833	0.894	0.872	0.856	0.856	0.878	0.850	0.950	**0.965**	0.850	0.871
LS	0.456	0.575	0.551	0.373	0.590	0.590	0.568	0.413	0.521	0.615	**0.669**
MI	0.510	0.390	0.500	0.51	0.610	0.500	0.590	0.535	0.515	0.640	**0.672**
NA	0.860	0.850	0.883	0.889	0.883	0.870	0.939	0.968	0.951	0.972	**0.972**
PD	0.973	0.939	0.977	0.978	0.977	0.948	0.980	0.985	0.983	**0.987**	0.946
PE	0.705	0.734	0.711	0.699	0.751	N/A	0.751	0.760	0.764	0.780	**0.812**
PS	0.104	0.151	0.151	0.11	0.298	0.190	0.175	0.299	0.295	0.309	**0.341**
RS	0.868	0.842	0.803	0.803	0.882	**0.934**	0.868	0.877	0.929	0.803	0.842
SCP1	0.771	0.765	0.775	0.874	0.782	0.710	0.652	0.835	0.829	0.898	**0.910**
SCP2	0.483	0.533	0.539	0.472	0.578	0.460	0.550	0.532	0.510	0.550	**0.589**
SWJ	0.200	0.333	0.200	0.067	0.533	0.333	0.400	0.383	0.383	0.467	**0.648**
UGL	0.881	0.869	0.903	0.891	0.906	0.916	0.894	**0.927**	0.926	0.850	0.859
Average accuracy	0.578	0.625	0.629	0.597	0.698	0.659	0.643	0.653	0.667	0.709	**0.740**
Total best accuracy	0	0	0	0	0	3	0	1	3	3	12
1-to-1-Wins	18	17	17	19	16	15	16	14	14	15	-
1-to-1-Losses/Draws	4	5	5	3	6	7	6	8	8	7	-

In summary, the overall classification performance of SAGoG across 22 datasets demonstrates that by learning the channel and instance correlations, SAGoG enhances the effectiveness of MTSC.

Based on the results in Table II, SAGoG (our model) and TodyNet, as graph-based models, achieve average accuracy rates of 0.740 and 0.709, respectively, which are notably higher than those of non-graph-based models (such as ED-INN with 0.578). SAGoG performs best on 12 datasets, while TodyNet excels on 3 datasets, indicating that graph-based modeling provides significant benefits in most cases. For example, on the FD dataset, the accuracies of SAGoG and TodyNet (0.683 and 0.627) are significantly better than those of non-graph-based models, suggesting that graph structures effectively capture relationships between variables, thereby improving performance. However, graph structures can also introduce noise, leading to poorer performance on certain datasets. For instance, on the AWR and EW datasets, the accuracies of SAGoG (0.946 and 0.815) are lower than those of WEASEL+MUSE (0.990 and 0.858), possibly due to inaccurate representation of variable relationships by the graph structure, which may cause interference. Nonetheless, the benefits of graph structures dominate on most datasets, resulting in a significant overall performance improvement.

As illustrated in Fig. 4, the proposed SAGoG model demonstrated significant performance advantages, achieving the lowest average rank of 3.659 across multiple datasets among all candidate models. The statistical significance of this ranking, determined utilizing the Wilcoxon signed-rank test with an alpha level of 0.05, further reinforces confidence in the robustness of SAGoG's performance. By adeptly encapsulating instance-level and channel-level dependencies, SAGoG has shown exceptional ability in capturing the complex dynamic patterns necessary for accurate classification.

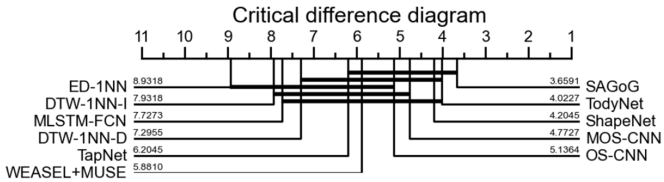

Fig. 4. Critical difference diagram on 22 UEA MTSC datasets.

Moreover, as shown in Table III, we conduct a focused comparison between SAGoG and three major categories of methods commonly used in time series modeling: graph neural network-based models, Transformer-based models, and large language model-based approaches. The analysis of experimental results on these three categories is as follows:

Comparison with Graph Neural Network-based Models (TISER-GCN and ForecastGrapher): Graph neural networks demonstrate certain advantages in multivariate time series classification. ForecastGrapher and TISER-GCN achieve average

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TABLE III
PERFORMANCE COMPARISON OF SAGOG AND MODELS BASED ON TRANSFORMER, GNN, AND LLM ON 22 UEA MTS DATASETS

Datasets	Prompt cast	TISER-GCN	Forecast Grapher	Auto former	Inf ormer	To dyNet	SAGoG
AWR	0.927	0.910	0.938	0.947	0.930	**0.987**	0.946
AF	0.452	0.393	0.410	0.333	0.267	**0.467**	0.430
DDG	0.638	0.600	0.610	0.520	0.540	0.580	**0.790**
EW	0.812	0.749	0.800	0.710	0.674	**0.840**	0.815
EP	0.925	0.910	**0.978**	0.942	0.949	0.971	0.928
FD	0.670	0.571	0.631	0.629	0.604	0.627	**0.683**
FM	0.559	0.470	0.512	0.520	0.530	0.570	**0.610**
HMD	0.572	0.425	0.390	0.378	0.338	**0.649**	0.541
HW	0.329	0.350	0.430	0.181	0.212	0.436	**0.643**
HB	0.720	0.680	0.644	0.698	**0.766**	0.756	0.761
LI	0.848	0.825	**0.900**	0.794	0.856	0.850	0.871
LS	0.602	0.500	0.449	0.625	0.634	0.615	**0.669**
MI	0.590	0.522	0.580	0.500	0.510	0.640	**0.672**
NA	0.945	0.810	0.882	0.933	0.956	0.972	**0.972**
PD	0.950	0.941	0.928	0.981	0.986	**0.987**	0.946
PE	0.792	0.690	0.736	0.717	0.775	0.780	**0.812**
PS	0.310	0.312	0.298	0.139	0.221	0.309	**0.341**
RS	0.821	0.748	0.800	0.809	0.783	0.803	**0.842**
SCP1	0.885	0.792	0.874	0.782	0.857	0.898	**0.910**
SCP2	0.505	0.510	0.564	0.544	0.578	0.550	**0.589**
SWJ	0.480	0.375	0.433	0.200	0.333	0.467	**0.648**
UGL	0.803	0.709	0.833	0.809	0.816	0.850	**0.859**
Avg	0.688	0.627	0.665	0.622	0.642	0.709	**0.740**
Best	0	0	2	0	1	5	14
W	19	22	20	19	19	15	-
L/D	3	0	2	3	3	7	-

accuracies of 0.665 and 0.627, respectively, outperforming some non-graph models. ForecastGrapher achieves a peak accuracy of 0.978 on the EP dataset, indicating its effectiveness in modeling structural dependencies. In contrast, TISER-GCN shows weaker overall stability. Although both models are competitive in specific tasks, their overall performance remains inferior to the proposed SAGoG model.

Comparison with Transformer-based Models (Autoformer and Informer): Autoformer and Informer achieve average accuracies of 0.622 and 0.642, respectively, which are lower than those of ForecastGrapher and SAGoG. While the Transformer architecture excels at capturing temporal dependencies in time series, its lack of explicit modeling of inter-variable structural relationships limits its performance in multivariate scenarios.

Comparison with the Large Language Model-based Method (PromptCast): PromptCast achieves an average accuracy of 0.688, but does not perform best on any dataset, indicating limited effectiveness in specific tasks. It performs well on datasets like AF, DDG, and PE, but underperforms on HW and SCP2, suggesting that improvements are needed in modeling temporal structure and inter-variable relationships. In contrast, SAGoG achieves the best results on 14 datasets, with an average accuracy of 0.740, demonstrating a clear advantage.

D. Ablation Study

To further understand the functions of main components in the SAGoG model, we conducted a series of ablation experiments. These experiments excluded specific graph structures during the classification modeling process to evaluate the impact of instance-level and channel-level dependencies, as well as other key features, on overall performance. The various variants of the SAGoG model are detailed as follows:

- *w/o Graph:* During classification, no graph structures are used to represent the time series. To facilitate classification, this model degrades into a method that combines GRU and FCN.
- *w/o ChGraph:* SAGoG without channel graph structures. In this variant, we only use instance graphs to represent the correlations between MTS instances. Correspondingly, the modules for channel graph structure evolving and channel graph pooling are also removed.
- *w/o InGraph:* SAGoG without instance graph structures. This variant excludes considerations of correlations between MTS instances and instead focuses exclusively on dynamic correlations between channels. After pooling the channel graphs to derive representations for each instance, classification is directly conducted.

Fig. 5 illustrates the classification results of three variants of the SAGoG model. Both axes in the figure represent classification accuracy. Specifically, the horizontal axis denotes the classification accuracy of each SAGoG variant, while the vertical axis represents the classification accuracy of the SAGoG model itself. Each subplot contains 22 scatter points representing experimental results on different UEA datasets. The diagonal line in the plot represents the line where horizontal and vertical axis values are equal. Points above this line indicate that the variant's classification performance is inferior to that of the SAGoG model.

Based on the results shown in Fig. 5, we conducted an analysis of three variants of the SAGoG model through ablation experiments. Subplot (a) displays the results of the SAGoG variant without using any graph structures, subplot (b) shows the results without using channel graph structures, and subplot (c) presents the results without using instance graph structures. The results clearly indicate that utilizing graph structures for modeling multivariate time series significantly enhances classification accuracy. In most cases, the scatter points lie above the diagonal line, indicating superior performance of the original SAGoG model compared to its variants without graph structures. Particularly noteworthy from the results in subplots (b) and (c) is the substantial improvement in classification performance achieved by incorporating channel graph and instance graph structures. This demonstrates SAGoG's capability to effectively capture dynamic dependencies between channels and correlations among instances in MTS data. However, there are instances where scatter points lie below the diagonal line in both subplots (b) and (c), suggesting weaker channel correlations or instance correlations in those specific datasets, potentially introducing noise. Nevertheless, the combined use of both types of graph structures for modeling MTS data in the SAGoG model has yielded favorable results, which explains the rationale behind employing this composite structure.

E. Inspection of Class Prototype

As shown in Fig. 6, this study employed the t-SNE method for visual analysis of time series embeddings learned by the

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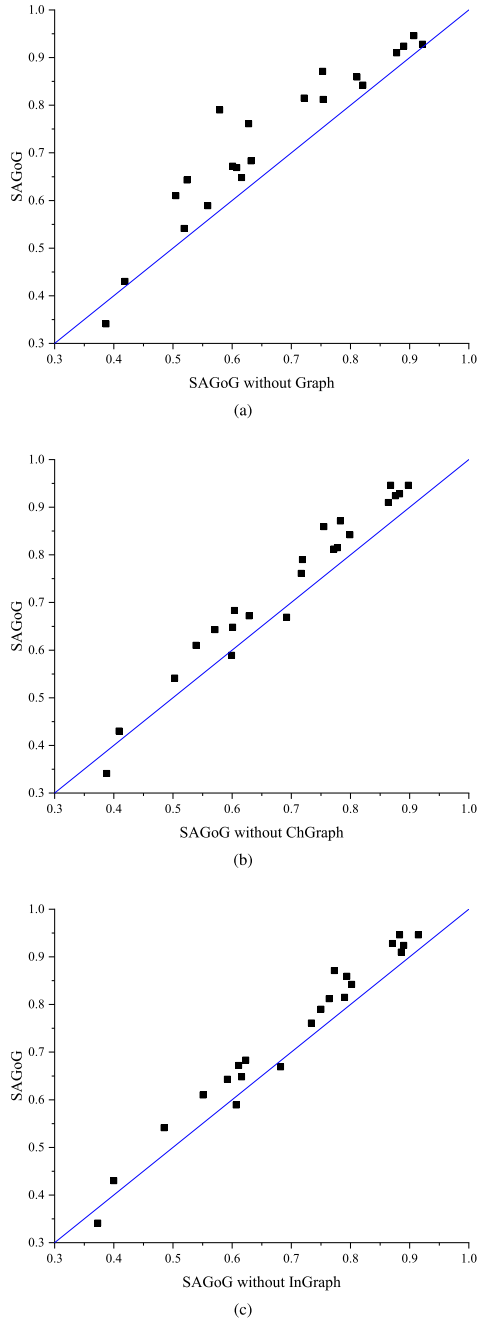

Fig. 5. Scatter plots of ablation results of the SAGoG model on 22 UEA dataset: SAGoG without graph mechanism (a), channel graphs (b), and SAGoG without instance graph (c).

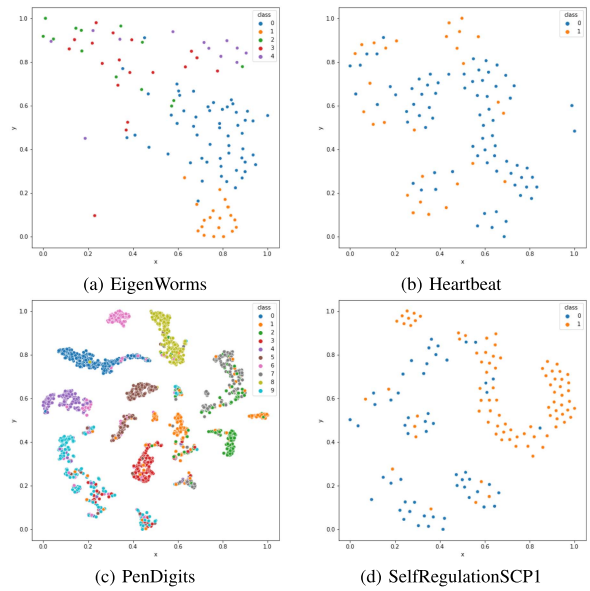

Fig. 6. The t-SNE visualization results of multivariate time series classification datasets EigenWorms, Heartbeat, PenDigits, and SelfRegulationSCP1.

SAGOG model on four datasets. Specifically, the visualization of the EigenWorms dataset revealed samples of different categories clustered in distinct colors, reflecting the model's effectiveness in preserving the intrinsic structure of the data, with clear boundaries between categories indicating a high degree of class discrimination. The visualization of the Heartbeat dataset showed clustering of different heartbeat patterns, which is crucial for the detection and prevention of heart disease, as abnormalities in these patterns may signal cardiac ailments. In the PenDigits dataset, the sample points of different handwritten digits also demonstrated significant clustering, indicating the model's capability to differentiate between the handwriting features of each digit. The visualization of the SelfRegulationSCP1 dataset depicted distinct clusters of brainwave patterns, showcasing the model's ability to capture individual brain activity patterns. Overall, these visualization results lead us to conclude that the SAGOG model can effectively distinguish and represent different categories of time series in multivariate time series classification tasks, which holds significant implications for subsequent research and application development in this field.

VI. Conclusion

In summary, we have introduced a novel MTSC framework called Similarity-Aware Graph of Graphs Neural Networks (SAGoG). Our approach effectively captures hidden dependencies among multiple instances of time series and dynamic channel correlations across different time intervals. By synthesizing channel-level and instance-level time series dependencies through channel graph structures and instance graph structures, respectively, and incorporating channel graph structure evolving and pooling mechanisms, SAGoG demonstrates strong performance in learning dependency features across variables and instances. Experimental results on the UEA dataset validate the superiority of our proposed model compared to traditional methods and deep learning-based models.

For future research, further investigations can be conducted to enhance the efficiency and scalability of channel graph representations and instance graph representations. Additionally, exploring the adaptability of SAGoG to different application domains and evaluating its robustness against noise or incomplete data could provide valuable insights.

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